



FINAL YEAR ENGINEERING PROJECT

Smart Medicine Availability & Intelligent Janaushadhi Recommendation System



PROJECT OVERVIEW

Project	Smart Medicine Availability & Intelligent Janaushadhi Recommendation System
Document	PROJECT OVERVIEW
Version	v1.0.0
Date	02 July 2026

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1. Project Introduction

This is a **Final Year Engineering Project** that builds a production-quality **React frontend** for a healthcare web application. The system helps ordinary citizens in India find affordable medicines, discover generic alternatives under the **Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)**, and locate nearby **Jan Aushadhi pharmacies** on an interactive map.

The application is designed for real-world use — it follows professional React architecture, uses an industry-standard design system, and is fully wired to integrate with a **FastAPI backend** whenever that backend is developed.

2. Problem Statement

Medicine prices in India can be extremely high for branded drugs. Most people are unaware that:

- Generic medicines with the **same active composition** are available at 50–90% lower cost
- Over **9,000 Jan Aushadhi stores** across India sell government-approved affordable generics
- Patients and doctors often do not know which generic alternative is available or where to get it

The problem this project solves:

> "A patient is prescribed an expensive branded medicine. They have no easy way to find if a cheaper generic version exists, what it is called, and where to buy it near them."

3. Objectives

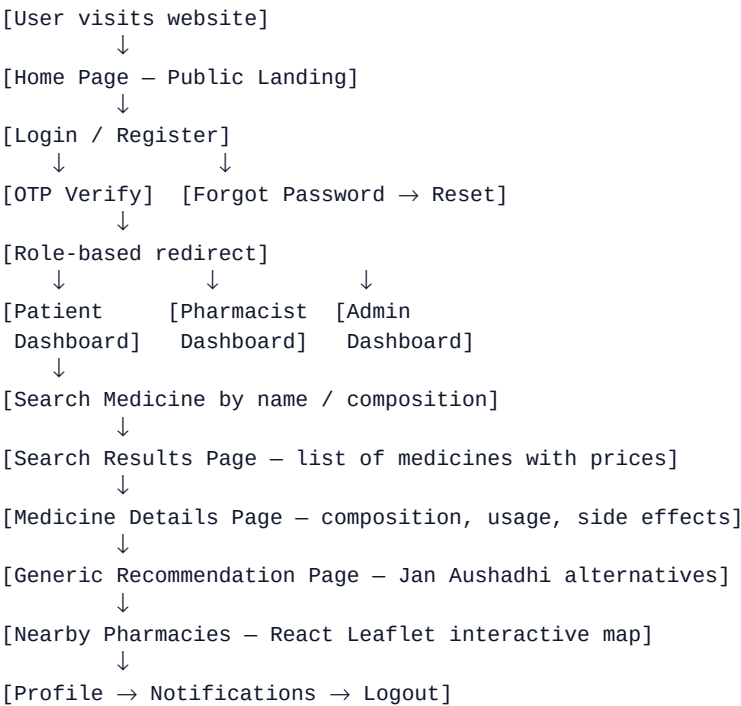
#	Objective
1	Allow users to search medicines by name or composition
2	Show generic / Jan Aushadhi alternatives with price comparison
3	Display nearby Jan Aushadhi pharmacies on an interactive map
4	Provide role-based dashboards for patients, pharmacists, and admins
5	Build a professional, accessible, and responsive UI
6	Architect the frontend so backend integration requires minimal effort

4. Target Users

Role	Who They Are	What They Do
Patient	Any Indian citizen	Search medicines, find generics, locate pharmacies

Role	Who They Are	What They Do
Doctor	Medical professional	Search medicines, recommend alternatives
Pharmacist	Pharmacy staff	Manage inventory, check stock
Admin	System administrator	Manage users, medicines, pharmacies, view analytics

5. Complete Frontend Workflow



6. Features

Patient Features

- Medicine search by name, brand, or composition
- Search results with smart badges (In Stock, Generic, Jan Aushadhi, Affordable)
- Medicine details with full information tabs
- Generic / Janaushadhi alternative recommendation
- Price comparison — branded vs generic
- Nearby pharmacies on interactive Leaflet map
- User dashboard with quick actions
- Notifications panel
- Profile management

Pharmacist Features

- Pharmacy dashboard with inventory summary
- Searchable, sortable, paginated inventory table
- Add / Edit medicine stock entries

- Prescriptions view (placeholder for future module)

Admin Features

- Admin dashboard with system metrics
- User management table
- Pharmacy management
- Medicine catalogue management
- Generic drug mapping
- Analytics and reports
- Notifications management
- Activity logs
- Role management
- System settings

System Features

- Light / dark theme toggle
- Offline detection (shows offline page)
- Session expiry detection
- Global error boundary (catches crashes)
- 404 Not Found page
- 403 Unauthorized page
- Lazy loading (all pages code-split)
- Responsive design (mobile, tablet, desktop)
- Keyboard accessible (ARIA labels, skip link)

7. Technologies Used

Technology	Version	Purpose
React	19.2.7	UI library
Vite	8.1.2	Build tool and dev server
React Router DOM	7.18.1	Client-side routing
Tailwind CSS	4.3.2	Utility-first CSS framework
TanStack React Query	5.101.2	Server-state caching
Axios	1.18.1	HTTP client for API calls
React Hook Form	7.80.0	Form state management
Zod	4.4.3	Schema-based form validation
React Leaflet	5.0.0	Interactive maps
Leaflet	1.9.4	Map engine
React Icons	5.7.0	Icon library (Herolcons, Material)
React Toastify	11.1.0	Toast notifications
ESLint	9.39.4	Code linting

8. Folder Structure

Frontend/	
docs/	← Project documentation (this folder)
public/	← Static assets
src/	
components/	← Reusable UI building blocks
cards/	← Card components
common/	← General utilities (Breadcrumb, SearchBar)
dialogs/	← Modal and ConfirmDialog
feedback/	← Spinner, Skeleton, EmptyState, ErrorState
forms/	← Input, Select, Checkbox, Toggle, etc.
layout/	← Container, PageHeader, SectionHeader
navigation/	← Navbar, Sidebar, Footer, TopBar
ui/	← Button, Badge, Avatar, Divider, IconButton
config/	← Axios, env variables, React Query client
constants/	← App constants, routes, nav config
contexts/	← React Context (Auth, Theme, User)
hooks/	← Custom React hooks
layouts/	← Page layout wrappers
pages/	← All application pages
routes/	← Router, ProtectedRoute, PublicRoute
services/	← Axios service layer for API calls
styles/	← Design tokens and theme config
utils/	← Pure utility functions
App.jsx	← Root application component
index.css	← Tailwind v4 CSS-first design tokens
main.jsx	← Entry point – mounts providers
index.html	← HTML shell
package.json	← Dependencies and scripts
vite.config.js	← Vite build configuration
eslint.config.js	← ESLint rules

9. Future Backend Integration

The frontend is **backend-ready**. Every API call is pre-wired in the service layer. When the **FastAPI backend** is developed, only the service files need to be updated — zero component changes required.

What to do	Where to do it
Set API base URL	.env → VITE_API_BASE_URL
Replace mock login	AuthContext.jsx → swap TODO comment
All medicine API calls	src/services/medicineService.js
All auth API calls	src/services/authService.js
All pharmacy API calls	src/services/pharmacyService.js
All user profile calls	src/services/userService.js
All inventory calls	src/services/inventoryService.js
Enable JWT session guard	App.jsx → set enabled: true in useSessionGuard

The backend API base URL is: <http://localhost:8000/api/v1> (configurable via .env).